MAX SEAL® Valves & Controls

PFA-Lined Ball Valves

Fluoro-Tek Series

Lined Valves Designed for High Corrosion and High Purity
Designed for Direct Mount Automation

Size Range:

1/4" - 16"

Pressure Rating:

All Sizes: 150 PSI MAWP

Materials:

Stainless Steel Body - A351 CF8M
PFA Inner Lining

Unique Design Features:

- Full-Bore Floating Ball Lined with Fluoroplastic
- Ball and Stem Cast as a Single Piece, Eliminating Rotation Angle Difference
- Anti-Blowout Stem Design
- Live-Loaded V-Ring PTFE Packing
- Spherical Body Cavity Split on the Centerline
- Multi-Channel Groove Design Locks the Lining to the Body
- Compact Body with High Strength
- 90° Operation by Handle, Gear or Actuator
- ISO 5211 "Direct Connect" Actuator Mounting
- Zero Seat Leakage
- Resistant to Strong Corrosive Media

Available with:

- V-Port Ball
- Vented Ball

Distributed by:



6" Model F15



Max-Seal brand name is cast onto every valve



1 1/2" Model F15

See Page 2 for design detail and structural advantages.

Max-Seal PFA-Lined Ball Valves are backed by the resources and experience of over 40 years of process valve and automation knowledge.

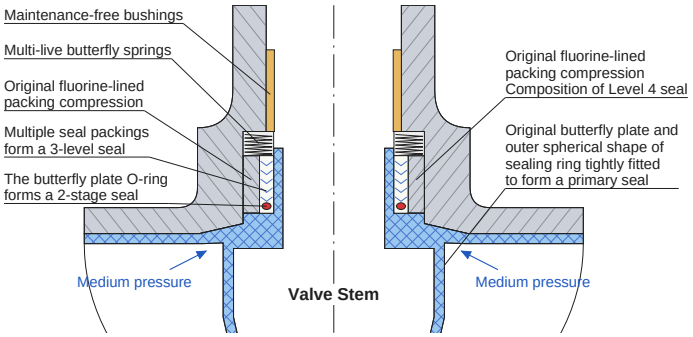
The Design

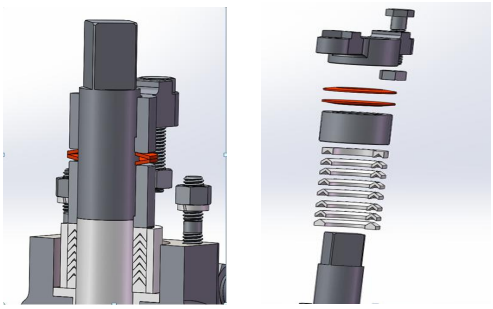
A special molding process is used to make the sealing surface very dense, and the combination of v-ring PTFE packing makes the valve achieve zero leakage. The ball and the valve stem are cast as a whole, singular piece, eliminating the possibility of the valve stem pushing out due to pressure changes and allowing for greater safety. It is constructed of an all-PFA lining, which is resistant to strong corrosive medias.

Features of the Fluorine-Lined Ball Valve

The valve is a full-bore floating ball type lined with fluoroplastic. The ball is integrated with the stem to eliminate the rotation angle difference (hysteresis). The stem has good anti-blow out performance, great sealing capability, essential safety, and reliability. The ball and the valve seats are tightly sealed, with zero seat leakage, and can use its plastic deformation to automatically compensate for the wear caused by severe media in order to achieve good sealing performance and long service life. In order to ensure the sealing at the valve stem is satisfactory, v-ring packing and packing gland are used in combination with springs that serve to both keep the valve grounded and provide live-loading to the packing. On the top of the bonnet is an ISO 5211-compliant mounting pad, designed to easily mount and install any gear or actuator, electric or pneumatic.

STEM SEALING SYSTEM

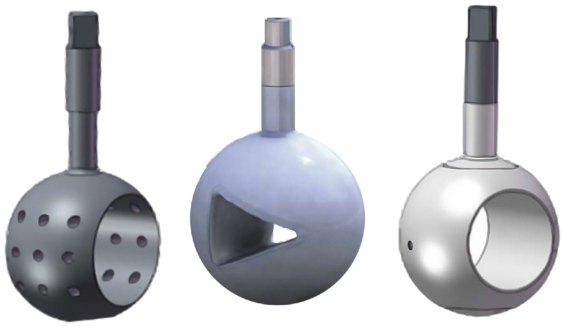


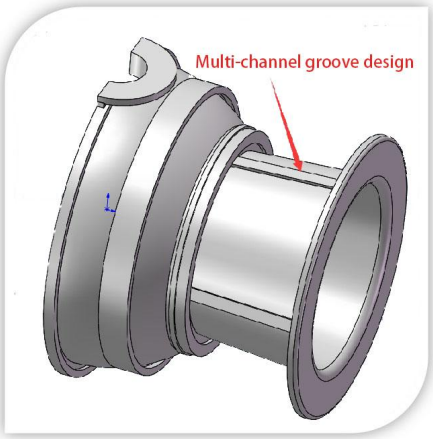


Structural Advantages

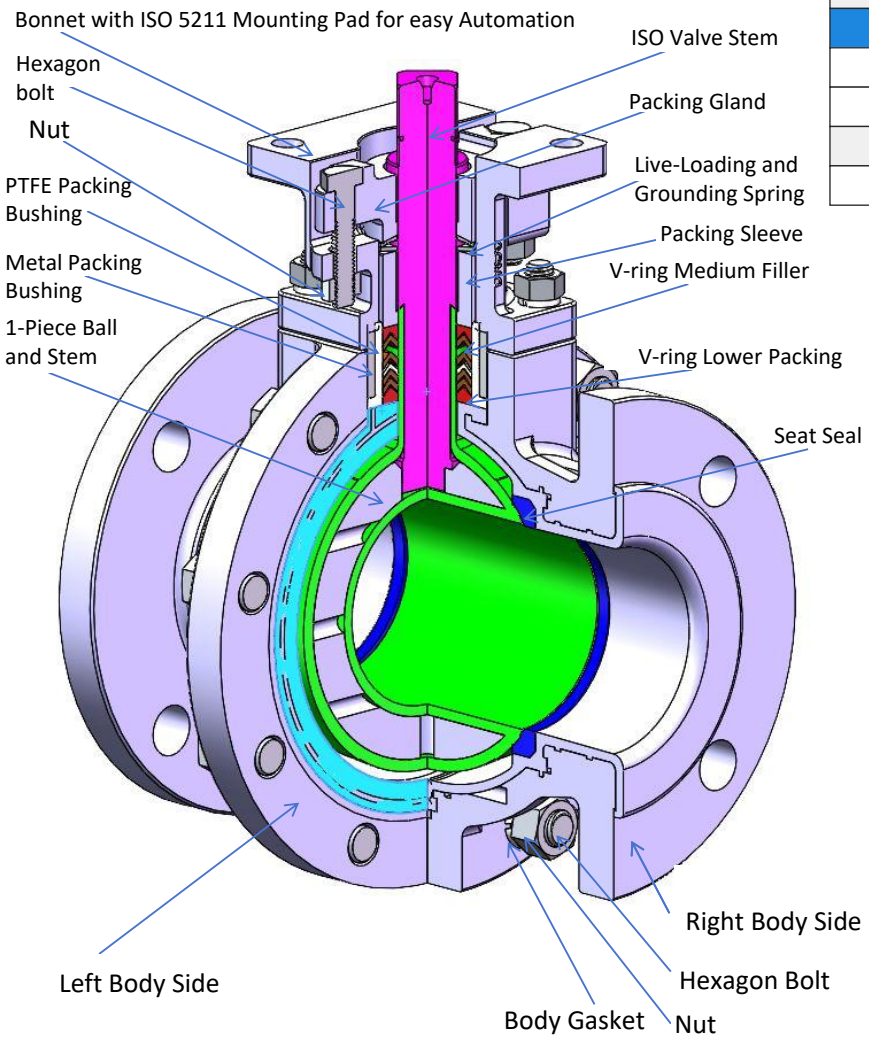
1. **Spherical body cavity design**
   The design of the valve body innovatively adopts the structure of the centerline to divide the left and right bodies so that the valve stem is not limited by length when it is loaded into the valve, and the valve's structure is more compact, with high strength.
2. **Live-Loaded Filler**
   The valve adopts butterfly spring live-loading preparation and low-leakage multi-layer v-ring packing design to solve any problem of packing dripping when the automatic control valve is opened frequently.
3. **The stem and spool are Integrally Formed**
   The stem and the ball core are integrally formed, the elongated valve stem has excellent transmission strength, and the ball core is machined with multiple process holes to make the fluoroplastic lining and steel tightly combined.
4. **Multi-Channel Groove Design**
   The inner lining is fixed with multiple hook grooves and threaded grooves so that the lining is combined with the metal shell. The valve cavity is designed with multiple grooves so that the fluorine lining layer is tightly engaged with the valve, preventing excessive wear of the lining.
5. **Full Port Design**
   The full port design ensures an unrestricted flow path for high flow rates with minimal pressure drop.

Available Ball Options





TECHNICAL SPECIFICATIONS



Performance Specification		
Pressure Rating	150 Class @ -20~100°F	
Temperature Range	-4~356°F, -20~180°C	

Testing Specification			
Pressure Test	Pressure		Test Medium
Shell Test	218 psig	1.5 MPa	Water
Seal Test	160 psig	1.1 MPa	
Air Seal Test	80 psig	0.6 MPa	Air

Bill of Materials

No.	Name	Material
1	Right Body	CF8M + PFA Lining
2	Left Body	CF8M + PFA Lining
3	Hexagon Bolt	A4-70
4	Nut	A4-70
5	Seat Seal	PTFE
6	Ball/Stem	316 + PFA Lining
7	PTFE Packing Bushing	PTFE
8	Metal Packing Bushing	316
9	V-ring Lower Packing	PTFE
10	V-ring Medium Filler	PTFE
11	V-ring Upper Packing	PTFE
12	Bonnet	CF8M
13	Packing Sleeves	316
14	Packing Gland	CF8M
15	Hexagon Bolt	A4-70
16	Nut	A4-70
17	Stud	A4-70
18	Nut	A4-70
19	Spring	50CRVA
20	Gasket	304
21	Handle	304

Classification of Lining Materials	
Lining Materials	Operating Temperature °C
FEP	-20 ~ 120
PTFE	-20 ~ 150
PFA	-20 ~ 180
Applicable Media	
Strong acid, strong alkali, strong oxidant, etc.	
Suitable Media	Corrosive Medium

Standards and Specifications:

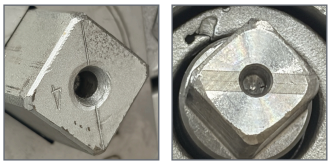
API 608 / JB/T 14312-2022 - PFA-Lined Design
ASME B16.10 - Face to Face
ASME B16.5 - Class 150 RF Flanges
ASTM B16.34 - Pressure/Temp Rating
API 598 - Testing Method

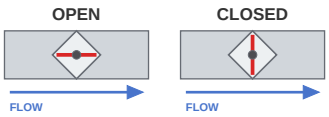
Valve Identification Tag



Stainless steel tag attached to every valve. Actual size: 60 mm x 13 mm.

Stem Position Indicator

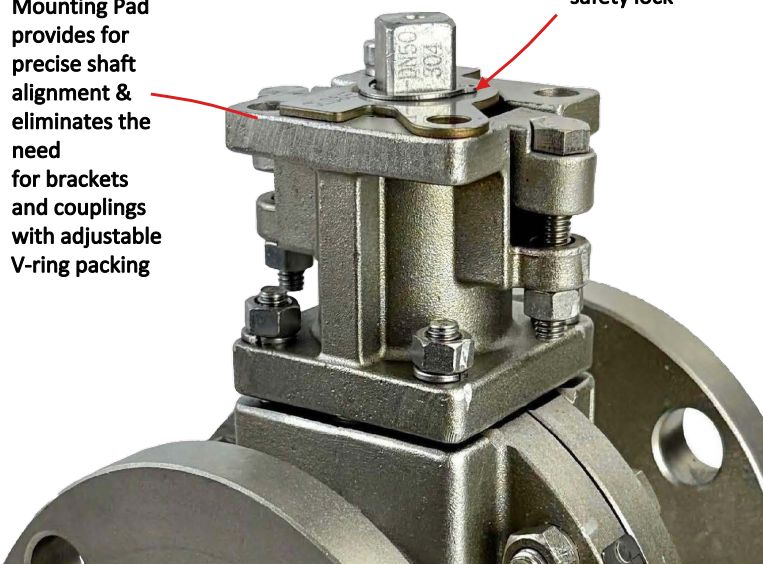




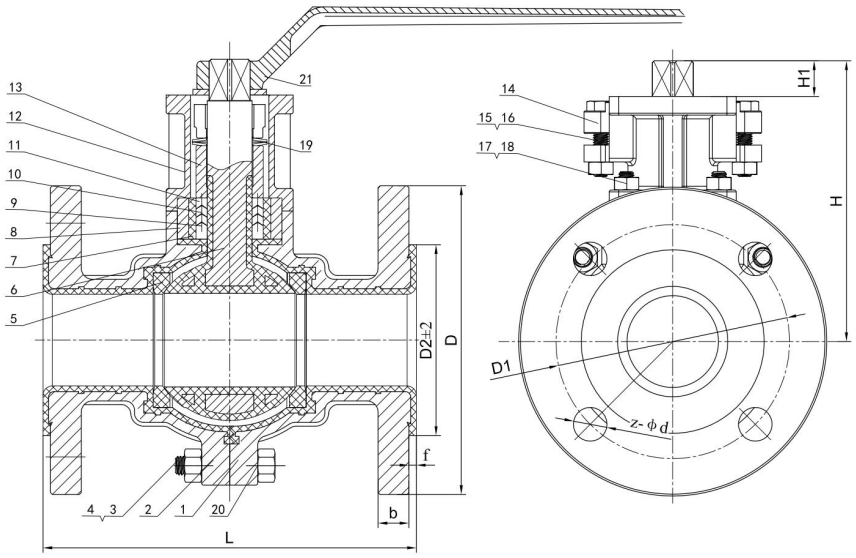
Indicator line on the stem top shows the ball port position.
Line in-line with the pipe = OPEN. Line across the pipe = CLOSED.

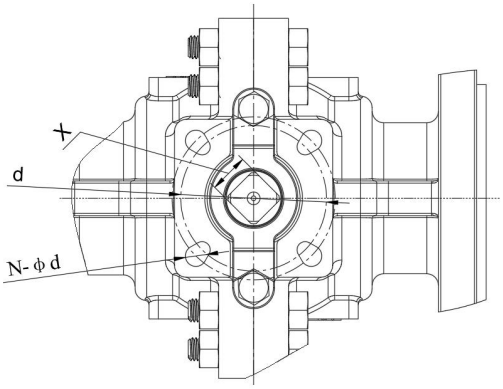
ISO 5211 Direct-connect Mounting Pad provides for precise shaft alignment & eliminates the need for brackets and couplings with adjustable V-ring packing

Hand-lever on ISO top flange for safety lock



ENGINEERING DETAILS & DIMENSIONS





Direct Connect Top Works

Size	1/2"	3/4"	1"	1 1/2"	2"	2 1/2"	3"	4"	6"
CV valve	19	45	88	242	425	783	1182	2130	3000

General Dimensions for PFA-Lined Class 150 RF Flanged Valve (≥10" sizes have a Fixed Trunnion Ball)

DN	NPS	L		D		D1		D2		z (#)	z - Ød		b		f		H		Weight	
mm	in	mm	in	mm	in	mm	in	mm (±2)	in (±0.08)		mm	in	mm	in	mm	in	mm	in	kg	lbs
15	1/2"	130	5.12	90	3.54	60.3	2.37	38	1.50	4	Ø16	Ø0.63	8	0.31	4	0.16	115	4.53	5.5	12
20	3/4"	140	5.51	100	3.94	69.9	2.75	46	1.81	4	Ø16	Ø0.63	8.9	0.35	4	0.16	115	4.53	7.4	16
25	1"	127	5.00	110	4.33	79.4	3.13	50.8	2.00	4	Ø16	Ø0.63	9.6	0.38	4	0.16	120	4.72	5.5	12
32	1-1/4"	140	5.51	115	4.53	88.9	3.50	63.5	2.50	4	Ø16	Ø0.63	11.2	0.44	4	0.16	128	5.04	7.4	16
40	1-1/2"	165	6.50	125	4.92	98.4	3.87	73	2.87	4	Ø16	Ø0.63	12.7	0.50	4	0.16	145	5.71	10.3	23
50	2"	178	7.01	150	5.91	120.7	4.75	92	3.62	4	Ø19	Ø0.75	14.3	0.56	4	0.16	150	5.91	11.5	25
65	2-1/2"	190	7.48	180	7.09	139.7	5.50	105	4.13	4	Ø19	Ø0.75	15.9	0.63	4	0.16	187	7.36	18.4	41
80	3"	203	7.99	190	7.48	152.4	6.00	126	4.96	4	Ø19	Ø0.75	17.5	0.69	4	0.16	200	7.87	23.3	51
100	4"	229	9.02	230	9.06	190.5	7.50	157	6.18	8	Ø19	Ø0.75	22.3	0.88	4	0.16	209	8.23	29.4	65
125	5"	356	14.02	255	10.04	215.9	8.50	186	7.32	8	Ø22.5	Ø0.89	22.3	0.88	5	0.20	245	9.65	38.8	86
150	6"	394	15.51	280	11.02	241.3	9.50	210	8.27	8	Ø22.5	Ø0.89	23.9	0.94	5	0.20	275	10.83	58.3	129
200	8"	457	17.99	340	13.39	295	11.61	265	10.43	8	Ø22.5	Ø0.89	27.0	1.06	6	0.24	345	13.58	113.5	250
250	10"	533	20.98	405	15.94	355	13.98	265	10.43	12	Ø25.5	Ø1.00	28.6	1.13	6	0.24	428	16.85	185	408
300	12"	610	24.02	460	18.11	410	16.14	265	10.43	12	Ø25.5	Ø1.00	30.2	1.19	6	0.24	458	18.03	363	800
350	14"	686	27.01	520	20.47	470	18.50	265	10.43	12	Ø25.5	Ø1.00	33.4	1.31	6	0.24	628	24.72	653	1440
400	16"	762	30.00	580	22.83	525	20.67	265	10.43	16	Ø28.5	Ø1.12	35.0	1.38	6	0.24	628	24.72	760	1676

Mounting Dimensions and Torques

ISO F-Pattern	d		N - Ød			X (45° offset square)		H1		Torque	
	mm	in	N (#)	mm	in	mm	in	mm	in	Nm	in-lbf
F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	25	221
F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	25	221
F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	30	266
F05 / F07	50 / 70	1.97 / 2.76	4	Ø8 / Ø10	Ø0.31 / Ø0.39	14 x 14	0.55 x 0.55	17	0.67	35	310
F07	70	2.76	4	Ø10 / Ø12	Ø0.39 / Ø0.47	17 x 17	0.67 x 0.67	19	0.75	50	443
F07 / F10	70 / 102	2.76 / 4.02	4	Ø10 / Ø12	Ø0.39 / Ø0.47	17 x 17	0.67 x 0.67	19	0.75	58	513
F10 / F12	102 / 125	4.02 / 4.92	4	Ø12 / Ø14	Ø0.47 / Ø0.55	22 x 22	0.87 x 0.87	23	0.91	86	761
F10 / F12	102 / 125	4.02 / 4.92	4	Ø12 / Ø14	Ø0.47 / Ø0.55	22 x 22	0.87 x 0.87	24	0.94	120	1062
F10 / F12	102 / 125	4.02 / 4.92	4	Ø12 / Ø14	Ø0.47 / Ø0.55	22 x 22	0.87 x 0.87	24	0.94	190	1682
F12 / F14	125 / 140	4.92 / 5.51	4	Ø14 / Ø18	Ø0.55 / Ø0.71	27 x 27	1.02 x 1.02	29	1.14	320	2832
F12 / F14	125 / 140	4.92 / 5.51	4	Ø14 / Ø18	Ø0.55 / Ø0.71	27 x 27	1.02 x 1.02	31	1.22	580	5133
F14 / F16	140 / 165	5.51 / 6.50	4	Ø18 / Ø22	Ø0.71 / Ø0.87	36 x 36	1.41 x 1.41	42	1.65	900	7965
F16 / F25	165 / 254	6.50 / 10.00	C/F			46 x 46	1.81 x 1.81	48	1.89	C/F	
F16 / F25	165 / 254	6.50 / 10.00	C/F			46 x 46	1.81 x 1.81	48	1.89	C/F	
F25 / F30	254 / 298	10.00 / 11.73	C/F			Ø70 x (2-20)	Ø70 x (2-20)	77	3.03	C/F	
F25 / F30	254 / 298	10.00 / 11.73	C/F			Ø70 x (2-20)	Ø70 x (2-20)	77	3.03	C/F	

VALVE INSPECTION / TESTING

Spark Testing
Ball: HG/T 4090-2009
Butterfly: GB/T 23711.1

ANTI-STATIC DEVICE

Press-fit metal disk ensures electrical continuity between the valve stem & body.

VACUUM RESISTANCE

Full

Max-Seal, Inc.
4815 West 5th St.
Lumberton, NC 28358

P.O. Box 1293
Lumberton, NC 28359
www.maxsealinc.com

910-738-2866
910-739-1733
sales@maxsealinc.com

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